



Neurobotics™

The Future is Here. Real. Now.

Neurobotics: The "Unlike LLMs" Technical Primer

The Probabilistic Flaw The contemporary artificial intelligence landscape is dominated by probabilistic software models and cloud-tethered large language models (LLMs). These systems excel at pattern approximation, generating outputs based on statistical likelihood rather than factual certainty. This introduces the "Hallucination Factor," making these models categorically unsafe for defense, aerospace, or medical-grade robotics.

The Deterministic Difference Neurobotics wholly rejects this probabilistic paradigm. Our architecture replaces stochastic, power-hungry autoregressive transformers with a hybrid Hyperdimensional Computing (HDC) and Symbolic Reasoning engine. By utilizing RISC-V and FPGA-inspired bare-metal frameworks, our systems execute logic with absolute mathematical certainty, ensuring 100% verifiable behavior with zero hallucination risk.

The Cloud Tether vs. Edge-Native Sovereignty Current LLM-based robot architectures rely on constant connectivity to high-power data centers. In austere or contested environments, a loss of signal results in a complete loss of autonomy. Furthermore, transmitting raw environmental data to third-party cloud servers breaks the chain of custody and violates zero-trust security frameworks.

Conversely, the Neurobotics architecture operates 100% off-grid. All cognitive vector space navigation occurs locally on the hardware; our systems require no external server connection to reason, act, or secure their environments.

Power and Physical Efficiency Traditional generative AI models require clusters of power-hungry GPUs, drawing kilowatts of power and demanding active liquid cooling. The Neurobotics architecture compresses complete situational awareness, rational logic, and real-time motor arbitration onto a single \$100mm $\times$ 100mm custom silicon board drawing less than 25W of total system power.

